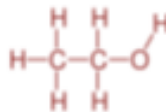
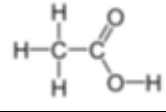
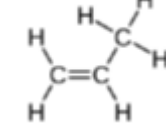
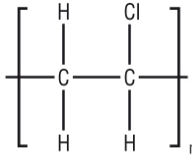
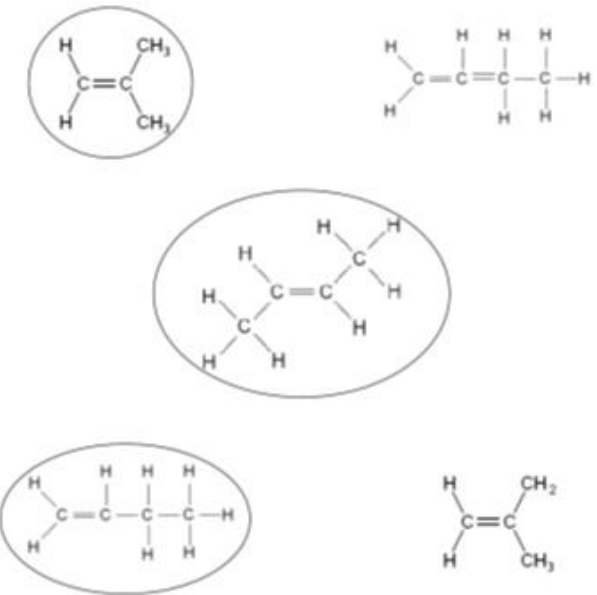
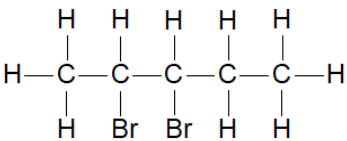


HIGHER TIER ONLY QUESTIONS

| Question |     |      |  | Marking details                                                                                                                                |                                  |                                                                                    |                          | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------|--------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                |                                  |                                                                                    |                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  | <b>Name</b>                                                                                                                                    | <b>Molecular formula</b>         | <b>Structure</b>                                                                   | <b>Homologous series</b> | 3               |     |     | 3     |       |      |
|          |     |      |  | ethanol                                                                                                                                        | C <sub>2</sub> H <sub>5</sub> OH |  | alcohols                 |                 |     |     |       |       |      |
|          |     |      |  | ethanoic acid                                                                                                                                  | CH <sub>3</sub> COOH             |  | carboxylic acids         |                 |     |     |       |       |      |
|          |     |      |  | propene                                                                                                                                        | C <sub>3</sub> H <sub>6</sub>    |  | alkenes                  |                 |     |     |       |       |      |
|          |     |      |  | award (1) for each correct answer                                                                                                              |                                  |                                                                                    |                          |                 |     |     |       |       |      |
|          |     | (ii) |  | C <sub>7</sub> H <sub>14</sub>                                                                                                                 |                                  |                                                                                    |                          |                 | 1   |     | 1     | 1     |      |
|          | (b) |      |  |  repeat unit (1)<br><br>n and side bonds (and brackets) (1) |                                  |                                                                                    |                          | 2               |     |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  |  |  <p>all three correct (2)<br/>any two correct (1)</p> | 2               |     |     | 2     |       |      |
|          | (d) |  |  |                                                      |                 | 1   |     | 1     |       |      |
|          |     |  |  | Question 3 total                                                                                                                        | 7               | 2   | 0   | 9     | 1     | 0    |